# Module 6: Overall Final Project

Vaultra Bank Project | Student Project Report

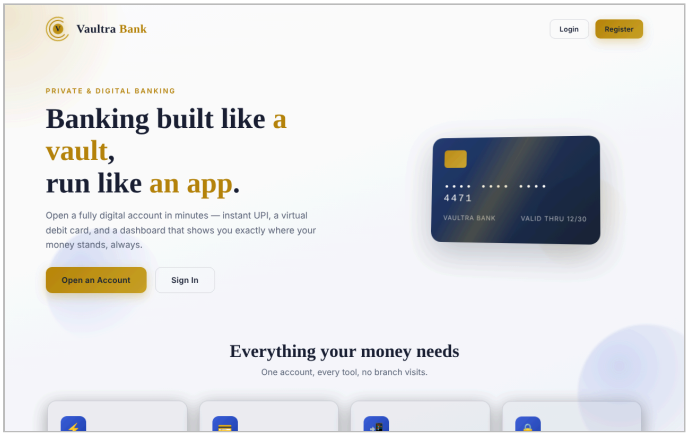
## 1. Aim

To combine all modules into one working project: **Vaultra Bank – Landing + Registration (Account Opening) + Login + Validation + Database + Dashboard**.

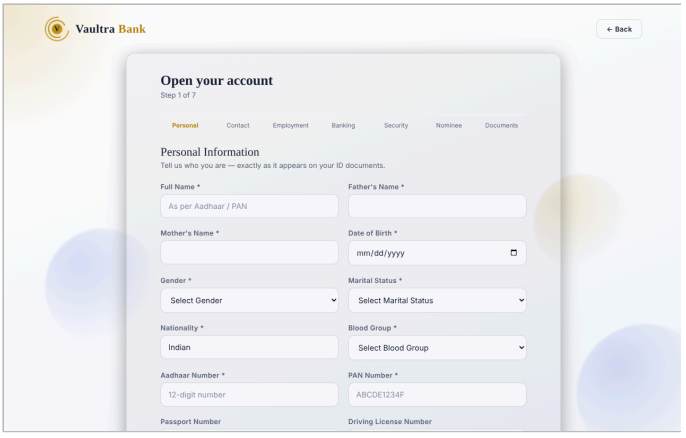
## 2. Description

The project is one HTML file (`vaultra_bank.html`). A new customer opens an account through a 7-step registration form (Personal, Contact, Employment, Banking, Security, Nominee, Documents). The script validates every step, generates a Customer ID, 16-digit Account Number, CIF number, IFSC code, UPI ID and a virtual debit card, and saves the customer in the browser database (`localStorage`). The customer then logs in and uses a dashboard with deposit, withdraw, transfer, transaction history, passbook download, card details, loan eligibility, fixed deposit, profile editing and settings.

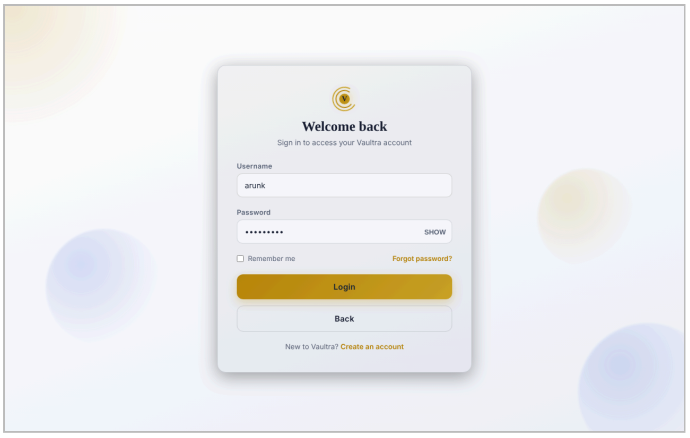
## 3. Working of the Project (screenshots)



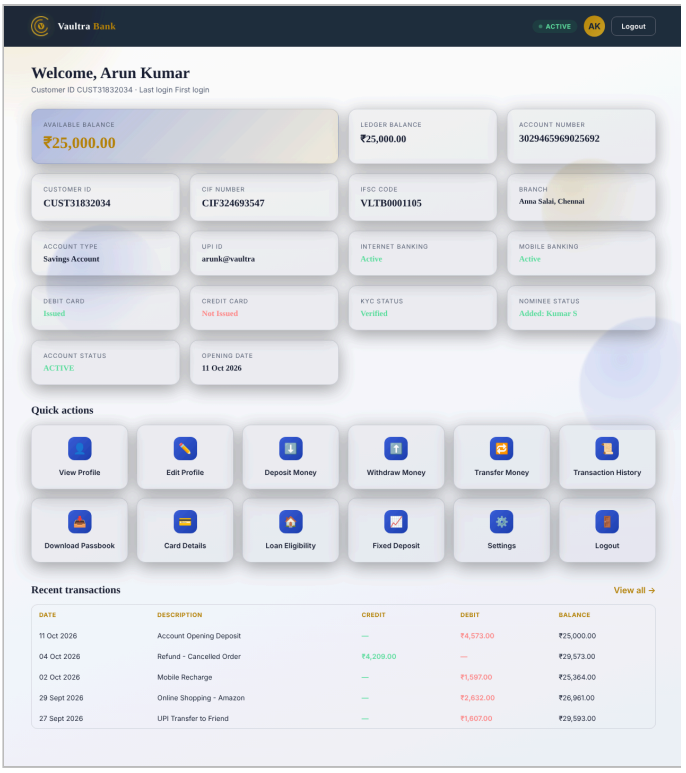

Step 1: Landing page



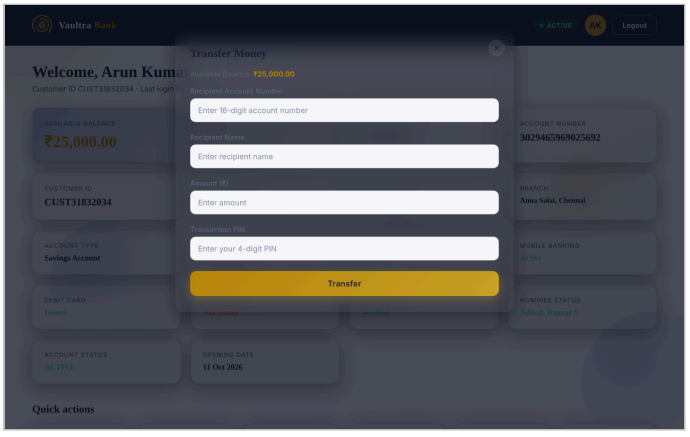

Step 2: Registration form (Step 1 of 7)



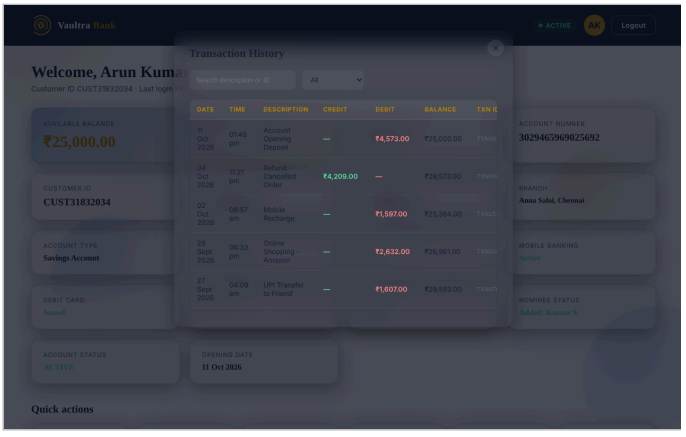

Step 3: Login page




Step 4: Dashboard after login (account details)




Step 5: Transfer Money (PIN protected)




Step 6: Transaction history with search & filter

**Sample output:** Registration with an initial deposit of ₹25,000 created Customer ID **CUST31832034**, Account Number **3029465969025692**, IFSC **VLTB0001105** and UPI ID **arunk@vaultra** (new random numbers are generated for every account).

## 4. How the Modules Are Joined

| Module | Role in the final project | Where in the code |
|---|---|---|
| 1. Bio data (HTML) | HTML structure, forms, tables and CSS styling used to design all pages | HTML + <style> |
| 2. Login page | Entry point; username + password checked against saved accounts | `#page-login`, `#loginForm` |
| 3. Registration page | 7-step account opening form with progress bar | `#page-register`, `FIELDS`, `renderSteps()` |
| 4. Validation scripting | Regex checks for email, mobile, Aadhaar, PAN, PIN code, password, names; required fields; deposit ≥ ₹500 | `PATTERNS`, `validateStep()` |
| 5. Database | Users, account data and transactions stored in browser `localStorage` (key `vaultra_users`) | `DB` object, `createAccount()` |

## 5. Program Flow

> Open page → Landing → Register (7 steps) → Each step valid? No: red message, stay / Yes: next step → Finish → Account created and saved → Login → Username + Password valid? No: error / Yes: Dashboard → Deposit / Withdraw / Transfer / FD → Transaction saved → Logout

## 6. How to Run

1. Save `vaultra_bank.html` on the computer and double-click it (any modern browser).
2. Click **Register** and complete all 7 steps (upload any image or PDF for each document and tick the three declarations).
3. After the success screen, log in with the username and password you created.
4. Use the quick action cards on the dashboard. Transfers need the 4-digit Transaction PIN set during registration.

## 7. Final Testing Summary

These tests were run automatically in a real browser (Chromium) against the submitted code.

| Area | Tests | Result |
|---|---|---|
| Registration — empty step, Aadhaar, PAN, mobile, email, deposit, password rules, documents, full valid signup | 12 | All passed |
| Login — valid login, wrong password, unknown user, empty fields, logout | 5 | All passed |
| Banking — deposit, withdraw, over-withdrawal, transfer PIN, fixed deposit | 6 | All passed |
| **Total** | **23** | **All 23 passed** |

## 8. Project Summary

| Item | Details |
|---|---|
| Project name | Vaultra Bank – Digital Banking Portal |
| Technologies | HTML5, CSS3, JavaScript, browser localStorage |
| Users | Any customer who completes registration |
| Output | Customer ID, Account Number, IFSC, UPI ID, virtual card, dashboard, transaction history, downloadable passbook |
| Limitations | Demo only: data and passwords are stored in the browser (not encrypted) and no real money moves; uploaded documents are not actually stored |
| Future improvements | Server-side login with hashed passwords, a real database (MySQL/SQLite), OTP verification, e-mail confirmation, admin page for KYC approval |

## 9. Complete Source Code: vaultra_bank.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Vaultra Bank | Banking, Simplified</title>
<style>
:root{
  --navy:#182338;
  --deep-blue:#0f2557;
  --royal:#1e40af;
  --royal-light:#3b5fd9;
  --gold:#c9a227;
  --gold-light:#b8860b;
  --white:#ffffff;
  --ink:#0a1628;
  --glass:rgba(15,25,50,0.045);
  --glass-border:rgba(15,25,50,0.14);
  --text-light:#1a2236;
  --text-muted:#5b6478;
  --success:#1fa864;
  --danger:#d64545;
  --radius:16px;
  --font-display: Georgia, 'Times New Roman', serif;
  --font-body: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
}
*{margin:0;padding:0;box-sizing:border-box;-webkit-tap-highlight-color:transparent;}
html{scroll-behavior:smooth;}
body{
  font-family:var(--font-body);
  background:linear-gradient(160deg,#ffffff 0%,#f7f8fc 45%,#eef1f8 100%);
  color:var(--text-light);
  min-height:100vh;
  overflow-x:hidden;
  position:relative;
}
::selection{background:var(--gold);color:var(--navy);}
:focus-visible{outline:2px solid var(--gold-light);outline-offset:2px;}

/* Floating background shapes */
.bg-shapes{position:fixed;inset:0;overflow:hidden;z-index:0;pointer-events:none;}
.shape{position:absolute;border-radius:50%;filter:blur(2px);opacity:.18;animation:floatShape 18s ease-in-out infinite;}
.shape.s1{width:340px;height:340px;background:radial-gradient(circle at 30% 30%,var(--gold-light),transparent 70%);top:-80px;le
ft:-100px;animation-duration:22s;}
.shape.s2{width:260px;height:260px;background:radial-gradient(circle at 30% 30%,var(--royal-light),transparent 70%);bottom:-60p
x;right:-60px;animation-duration:26s;animation-delay:2s;}
.shape.s3{width:180px;height:180px;background:radial-gradient(circle at 30% 30%,var(--gold),transparent 70%);top:40%;right:8%;a
nimation-duration:20s;animation-delay:1s;}
.shape.s4{width:220px;height:220px;background:radial-gradient(circle at 30% 30%,var(--royal-light),transparent 70%);bottom:20%;
left:6%;animation-duration:24s;animation-delay:3s;}
@keyframes floatShape{
  0%,100%{transform:translate(0,0) scale(1);}
  33%{transform:translate(30px,-40px) scale(1.08);}
  66%{transform:translate(-25px,25px) scale(.95);}
```

```
}

.app{position:relative;z-index:1;min-height:100vh;}

/* Page transition system */
.page{display:none;min-height:100vh;}
.page.active{display:block;animation:pageFadeSlide .55s ease both;}
@keyframes pageFadeSlide{
  from{opacity:0;transform:translateY(24px) scale(.99);}
  to{opacity:1;transform:translateY(0) scale(1);}
}
.page.leaving{animation:pageFadeOut .35s ease both;}
@keyframes pageFadeOut{
  from{opacity:1;transform:translateY(0);}
  to{opacity:0;transform:translateY(-16px);}
}

/* Glass panel */
.glass{
  background:var(--glass);
  border:1px solid var(--glass-border);
  backdrop-filter:blur(18px);
  -webkit-backdrop-filter:blur(18px);
  border-radius:var(--radius);
  box-shadow:0 8px 32px rgba(0,0,0,.35);
}

/* Buttons */
.btn{
  position:relative;overflow:hidden;
  display:inline-flex;align-items:center;justify-content:center;gap:8px;
  padding:14px 30px;border-radius:12px;border:none;cursor:pointer;
  font-family:var(--font-body);font-size:15px;font-weight:600;letter-spacing:.3px;
  transition:transform .2s ease, box-shadow .2s ease, background .2s ease;
  white-space:nowrap;
}
.btn:active{transform:scale(.97);}
.btn-gold{background:linear-gradient(135deg,var(--gold-light),var(--gold));color:var(--navy);box-shadow:0 6px 18px rgba(201,16
2,39,.35);}
.btn-gold:hover{box-shadow:0 10px 26px rgba(201,162,39,.5);transform:translateY(-2px);}
.btn-outline{background:transparent;border:1.5px solid var(--glass-border);color:var(--text-light);}
.btn-outline:hover{background:rgba(255,255,255,.08);border-color:var(--gold-light);}
.btn-royal{background:linear-gradient(135deg,var(--royal-light),var(--royal));color:#fff;box-shadow:0 6px 18px rgba(30,64,175,.
4);}
.btn-royal:hover{box-shadow:0 10px 26px rgba(30,64,175,.55);transform:translateY(-2px);}
.btn-danger{background:linear-gradient(135deg,#e05d5d,var(--danger));color:#fff;}
.btn-sm{padding:9px 18px;font-size:13px;border-radius:9px;}
.btn-block{width:100%;}
.btn[disabled]{opacity:.5;cursor:not-allowed;transform:none !important;}
.ripple{position:absolute;border-radius:50%;background:rgba(255,255,255,.55);transform:scale(0);animation:rippleAnim .6s ease-o
ut;pointer-events:none;}
@keyframes rippleAnim{to{transform:scale(3);opacity:0;}}

/* Landing */
.landing-nav{display:flex;align-items:center;justify-content:space-between;padding:22px 6%;}
.brand{display:flex;align-items:center;gap:12px;}
.logo{width:46px;height:46px;position:relative;}
.logo-ring{position:absolute;inset:0;border-radius:50%;border:2px solid var(--gold);border-top-color:transparent;animation:spin
4s linear infinite;}
.logo-ring.r2{inset:6px;border-color:var(--gold-light);border-bottom-color:transparent;animation:spin 3s linear infinite revers
e;}
.logo-core{position:absolute;inset:13px;background:radial-gradient(circle,var(--gold-light),var(--gold));border-radius:50%;disp
lay:flex;align-items:center;justify-content:center;font-family:var(--font-display);font-weight:700;color:var(--navy);font-size:
14px;box-shadow:0 0 16px rgba(232,199,102,.7);animation:pulseGlow 2.4s ease-in-out infinite;}
@keyframes spin{to{transform:rotate(360deg);}}
@keyframes pulseGlow{0%,100%{box-shadow:0 0 10px rgba(232,199,102,.5);}50%{box-shadow:0 0 22px rgba(232,199,102,.9);}}
.brand-name{font-family:var(--font-display);font-size:22px;font-weight:700;letter-spacing:.5px;}
.brand-name span{color:var(--gold-light);}
.nav-links{display:flex;gap:12px;}

.hero{padding:60px 6% 40px;display:flex;flex-wrap:wrap;align-items:center;gap:48px;}
.hero-text{flex:1 1 420px;animation:fadeSlideUp .8s ease both;}
@keyframes fadeSlideUp{from{opacity:0;transform:translateY(30px);}to{opacity:1;transform:translateY(0);}}
.eyebrow{display:inline-block;color:var(--gold-light);font-size:13px;letter-spacing:2.5px;text-transform:uppercase;margin-botto
m:14px;font-weight:600;}
.hero-text h1{font-family:var(--font-display);font-size:clamp(32px,5vw,54px);line-height:1.12;margin-bottom:18px;}
.hero-text h1 em{font-style:normal;color:var(--gold-light);}
.hero-text p{color:var(--text-muted);font-size:17px;max-width:520px;margin-bottom:28px;line-height:1.6;}
.hero-actions{display:flex;gap:16px;flex-wrap:wrap;}
.hero-visual{flex:1 1 320px;display:flex;justify-content:center;animation:fadeSlideUp 1s ease .1s both;}
.vault-card{width:320px;height:200px;border-radius:20px;position:relative;background:linear-gradient(135deg,#16306b,#0a1628);bo
x-shadow:0 20px 60px rgba(10,22,40,.25);padding:24px;overflow:hidden;transform:perspective(800px) rotateY(-8deg) rotateX(4deg);
transition:transform .4s ease;color:#fff;}
.vault-card:hover{transform:perspective(800px) rotateY(0) rotateX(0);}
.vault-card::before{content:"";position:absolute;top:-40%;left:-20%;width:180%;height:180%;background:linear-gradient(120deg,tr
ansparent 40%,rgba(201,162,39,.25) 50%,transparent 60%);animation:shimmer 3.5s linear infinite;}
@keyframes shimmer{0%{transform:translateX(-30%);}100%{transform:translateX(30%);}}
.vault-chip{width:44px;height:34px;border-radius:6px;background:linear-gradient(135deg,var(--gold-light),var(--gold));margin-bo
ttom:26px;}
.vault-num{font-size:19px;letter-spacing:3px;font-family:'Courier New',monospace;margin-bottom:18px;color:#fff;}
```

```
.vault-foot{display:flex;justify-content:space-between;font-size:12px;color:rgba(255,255,255,.65);letter-spacing:1px;}

.features{padding:50px 6% 80px;}
.section-title{text-align:center;font-family:var(--font-display);font-size:clamp(24px,3.4vw,34px);margin-bottom:10px;}
.section-sub{text-align:center;color:var(--text-muted);margin-bottom:44px;}
.feature-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(230px,1fr));gap:22px;max-width:1120px;margin:0 auto;}
.feature-card{padding:28px 24px;transition:transform .35s ease, box-shadow .35s ease, border-color .35s ease;animation:fadeSlid
eUp .7s ease both;}
.feature-card:hover{transform:translateY(-8px);box-shadow:0 16px 40px rgba(0,0,0,.45);border-color:rgba(232,199,102,.5);}
.feature-icon{width:48px;height:48px;border-radius:12px;background:linear-gradient(135deg,var(--royal-light),var(--royal));disp
lay:flex;align-items:center;justify-content:center;font-size:22px;margin-bottom:16px;}
.feature-card h3{font-size:17px;margin-bottom:8px;}
.feature-card p{font-size:14px;color:var(--text-muted);line-height:1.5;}

footer.site-footer{text-align:center;padding:30px 6% 46px;color:var(--text-muted);font-size:13px;}

/* Auth pages */
.auth-wrap{min-height:100vh;display:flex;align-items:center;justify-content:center;padding:40px 5%;}
.auth-card{width:100%;max-width:480px;padding:40px 36px;animation:zoomIn .5s ease both;}
@keyframes zoomIn{from{opacity:0;transform:scale(.92);}to{opacity:1;transform:scale(1);}}
.auth-head{text-align:center;margin-bottom:28px;}
.auth-head h2{font-family:var(--font-display);font-size:26px;margin:14px 0 6px;}
.auth-head p{color:var(--text-muted);font-size:14px;}

.field{position:relative;margin-bottom:20px;}
.field label{display:block;font-size:13px;color:var(--text-muted);margin-bottom:6px;font-weight:600;letter-spacing:.3px;}
.field input, .field select, .field textarea{
  width:100%;padding:13px 14px;border-radius:10px;border:1.5px solid var(--glass-border);
  background:#f6f7fb;color:var(--text-light);font-size:14.5px;font-family:var(--font-body);
  transition:border-color .25s ease, box-shadow .25s ease, background .25s ease;
}
.field input::placeholder{color:#8992a8;}
.field input:focus, .field select:focus, .field textarea:focus{
  outline:none;border-color:var(--gold-light);background:#ffffff;
  box-shadow:0 0 0 4px rgba(201,162,39,.15);
}
.field select option{background:#ffffff;color:#1a2236;}
.field.error input, .field.error select{border-color:var(--danger);box-shadow:0 0 0 4px rgba(214,69,69,.15);}
.error-msg{color:#ff8a8a;font-size:12px;margin-top:6px;display:none;animation:shake .4s ease;}
.field.error .error-msg{display:block;}
@keyframes shake{0%,100%{transform:translateX(0);}20%{transform:translateX(-6px);}40%{transform:translateX(6px);}60%{transform:
translateX(-4px);}80%{transform:translateX(4px);}}
.pw-wrap{position:relative;}
.pw-toggle{position:absolute;right:12px;top:38px;background:none;border:none;color:var(--text-muted);cursor:pointer;font-size:1
3px;font-weight:600;}
.check-row{display:flex;align-items:flex-start;gap:10px;margin-bottom:14px;font-size:13.5px;color:var(--text-muted);}
.check-row input{margin-top:3px;accent-color:var(--gold);}
.row-between{display:flex;justify-content:space-between;align-items:center;font-size:13px;margin-bottom:22px;color:var(--text-m
uted);}
.link{color:var(--gold-light);cursor:pointer;text-decoration:none;font-weight:600;}
.link:hover{text-decoration:underline;}
.auth-foot{text-align:center;margin-top:20px;font-size:13.5px;color:var(--text-muted);}

/* Registration multi-step */
.reg-wrap{max-width:820px;margin:0 auto;padding:50px 5% 80px;}
.progress-track{position:relative;height:6px;background:rgba(255,255,255,.1);border-radius:10px;margin:26px 0 10px;overflow:hid
den;}
.progress-fill{height:100%;width:0%;background:linear-gradient(90deg,var(--gold),var(--gold-light));border-radius:10px;transiti
on:width .5s cubic-bezier(.4,0,.2,1);}
.step-labels{display:flex;justify-content:space-between;flex-wrap:wrap;gap:6px;margin-bottom:30px;font-size:11.5px;color:var(--
text-muted);}
.step-labels span{flex:1;text-align:center;min-width:60px;transition:color .3s;}
.step-labels span.active{color:var(--gold-light);font-weight:700;}
.step-labels span.done{color:var(--success);}
.form-step{display:none;animation:slideFadeIn .45s ease both;}
.form-step.active{display:block;}
@keyframes slideFadeIn{from{opacity:0;transform:translateX(24px);}to{opacity:1;transform:translateX(0);}}
.step-title{font-family:var(--font-display);font-size:22px;margin-bottom:4px;}
.step-desc{color:var(--text-muted);font-size:13.5px;margin-bottom:24px;}
.grid2{display:grid;grid-template-columns:1fr 1fr;gap:0 18px;}
@media (max-width:640px){.grid2{grid-template-columns:1fr;}}
.check-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(180px,1fr));gap:10px 16px;margin-bottom:10px;}
.toggle-item{display:flex;align-items:center;justify-content:space-between;padding:12px 14px;border-radius:10px;background:rgba
(255,255,255,.04);border:1px solid var(--glass-border);}
.switch{position:relative;width:42px;height:24px;flex-shrink:0;}
.switch input{opacity:0;width:0;height:0;}
.slider-toggle{position:absolute;cursor:pointer;inset:0;background:rgba(255,255,255,.2);border-radius:24px;transition:.3s;}
.slider-toggle::before{content:"";position:absolute;height:18px;width:18px;left:3px;top:3px;background:#fff;border-radius:50%;t
ransition:.3s;}
.switch input:checked + .slider-toggle{background:linear-gradient(135deg,var(--gold-light),var(--gold));}
.switch input:checked + .slider-toggle::before{transform:translateX(18px);}
.upload-box{border:2px dashed var(--glass-border);border-radius:12px;padding:22px;text-align:center;cursor:pointer;transition:b
order-color .3s, background .3s;position:relative;margin-bottom:18px;}
.upload-box:hover{border-color:var(--gold-light);background:rgba(255,255,255,.04);}
.upload-box input{position:absolute;inset:0;opacity:0;cursor:pointer;}
.upload-box .fname{margin-top:8px;font-size:12.5px;color:var(--gold-light);font-weight:600;}
.upload-box.error{border-color:var(--danger);}
.check-row.error{background:rgba(214,69,69,.08);border-radius:8px;}
.step-nav{display:flex;justify-content:space-between;margin-top:32px;gap:14px;}
```

```
/* Verification overlay */
.overlay{position:fixed;inset:0;background:rgba(6,12,30,.94);z-index:1000;display:none;align-items:center;justify-content:cente
r;padding:20px;}
.overlay.active{display:flex;animation:fadeIn .3s ease;}
@keyframes fadeIn{from{opacity:0;}to{opacity:1;}}
.verify-box{width:100%;max-width:440px;text-align:center;padding:44px 32px;}
.spinner{width:70px;height:70px;border-radius:50%;border:4px solid rgba(255,255,255,.15);border-top-color:var(--gold-light);mar
gin:0 auto 24px;animation:spin 1s linear infinite;}
.verify-title{font-family:var(--font-display);font-size:20px;margin-bottom:22px;}
.verify-list{text-align:left;margin:0 auto 24px;max-width:320px;}
.verify-item{display:flex;align-items:center;gap:10px;padding:8px 0;font-size:13.5px;color:var(--text-muted);opacity:.4;transit
ion:opacity .3s;}
.verify-item.done{opacity:1;color:var(--text-light);}
.verify-item .tick{width:18px;height:18px;border-radius:50%;background:var(--glass-border);flex-shrink:0;display:flex;align-ite
ms:center;justify-content:center;font-size:11px;transition:background .3s;}
.verify-item.done .tick{background:var(--success);}
.verify-item.done .tick::after{content:"✓";}
.overlay .progress-track{margin:0 0 6px;}
.progress-pct{font-size:12px;color:var(--text-muted);margin-bottom:10px;}
.success-check{width:84px;height:84px;border-radius:50%;background:radial-gradient(circle,rgba(31,168,100,.25),transparent 7
0%);display:flex;align-items:center;justify-content:center;margin:0 auto 18px;}
.success-check .circle{width:64px;height:64px;border-radius:50%;background:var(--success);display:flex;align-items:center;justi
fy-content:center;animation:bounceIn .6s cubic-bezier(.34,1.56,.64,1) both;}
.success-check svg{width:32px;height:32px;}
.success-check path{stroke-dasharray:40;stroke-dashoffset:40;animation:drawCheck .5s ease .3s forwards;}
@keyframes drawCheck{to{stroke-dashoffset:0;}}
@keyframes bounceIn{0%{transform:scale(0);}60%{transform:scale(1.15);}100%{transform:scale(1);}}
.success-title{font-family:var(--font-display);font-size:24px;margin-bottom:8px;}
.success-sub{color:var(--text-muted);font-size:14px;}

/* Confetti */
.confetti-piece{position:fixed;top:-20px;width:9px;height:14px;z-index:1200;pointer-events:none;animation:confettiFall linear f
orwards;}
@keyframes confettiFall{to{transform:translateY(105vh) rotate(720deg);opacity:.9;}}

/* Dashboard */
.dash-nav{display:flex;align-items:center;justify-content:space-between;padding:18px 4%;position:sticky;top:0;z-index:50;backgr
ound:rgba(10,22,40,.9);backdrop-filter:blur(14px);border-bottom:1px solid rgba(255,255,255,.08);color:#fff;}
.dash-nav .brand-name{color:#fff;}
.dash-nav .btn-outline{color:#fff;border-color:rgba(255,255,255,.3);}
.dash-nav .btn-outline:hover{background:rgba(255,255,255,.1);}
.dash-user{display:flex;align-items:center;gap:12px;}
.avatar{width:40px;height:40px;border-radius:50%;background:linear-gradient(135deg,var(--gold-light),var(--gold));display:flex;
align-items:center;justify-content:center;font-weight:700;color:var(--navy);overflow:hidden;}
.avatar img{width:100%;height:100%;object-fit:cover;}
.dash-body{padding:32px 4% 70px;max-width:1280px;margin:0 auto;}
.welcome-bar{margin-bottom:28px;animation:fadeSlideUp .6s ease both;}
.welcome-bar h1{font-family:var(--font-display);font-size:clamp(22px,3vw,30px);margin-bottom:6px;}
.welcome-bar p{color:var(--text-muted);font-size:14px;}
.status-pill{display:inline-flex;align-items:center;gap:6px;padding:5px 12px;border-radius:20px;background:rgba(31,168,100,.1
5);color:#5fe0a0;font-size:12px;font-weight:700;letter-spacing:.5px;}
.status-pill .dot{width:7px;height:7px;border-radius:50%;background:#5fe0a0;animation:pulseDot 1.6s ease-in-out infinite;}
@keyframes pulseDot{0%,100%{opacity:1;}50%{opacity:.3;}}

.info-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(240px,1fr));gap:18px;margin-bottom:36px;}
.info-card{padding:22px;animation:fadeSlideUp .6s ease both;transition:transform .3s;}
.info-card:hover{transform:translateY(-4px);}
.info-card .label{font-size:11.5px;color:var(--text-muted);letter-spacing:1px;text-transform:uppercase;margin-bottom:8px;}
.info-card .value{font-size:19px;font-weight:700;font-family:var(--font-display);}
.info-card.balance{background:linear-gradient(135deg,rgba(30,64,175,.35),rgba(201,162,39,.15));grid-column:span 2;}
.info-card.balance .value{font-size:30px;color:var(--gold-light);}
@media (max-width:520px){.info-card.balance{grid-column:span 1;}}

.section-head{display:flex;align-items:center;justify-content:space-between;margin-bottom:16px;}
.section-head h2{font-family:var(--font-display);font-size:20px;}
.action-grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(180px,1fr));gap:16px;margin-bottom:40px;}
.action-card{padding:22px 18px;text-align:center;cursor:pointer;transition:transform .3s, box-shadow .3s, border-color .3s;anim
ation:fadeSlideUp .6s ease both;}
.action-card:hover{transform:translateY(-6px);box-shadow:0 14px 34px rgba(0,0,0,.4);border-color:rgba(232,199,102,.5);}
.action-icon{width:44px;height:44px;border-radius:12px;margin:0 auto 12px;background:linear-gradient(135deg,var(--royal-light),
var(--royal));display:flex;align-items:center;justify-content:center;font-size:20px;}
.action-card span{font-size:13.5px;font-weight:600;}

/* Modal */
.modal-overlay{position:fixed;inset:0;background:rgba(4,9,22,.72);z-index:900;display:none;align-items:flex-start;justify-conte
nt:center;padding:6vh 4% 4vh;overflow-y:auto;}
.modal-overlay.active{display:flex;animation:fadeIn .25s ease;}
.modal{width:100%;max-width:640px;padding:30px 28px;animation:zoomIn .35s ease both;position:relative;}
.modal-close{position:absolute;top:16px;right:16px;background:rgba(255,255,255,.08);border:none;color:var(--text-light);width:3
2px;height:32px;border-radius:50%;cursor:pointer;font-size:16px;transition:background .2s;}
.modal-close:hover{background:rgba(255,255,255,.18);}
.modal h2{font-family:var(--font-display);font-size:21px;margin-bottom:18px;padding-right:30px;}
.detail-row{display:flex;justify-content:space-between;padding:10px 0;border-bottom:1px dashed var(--glass-border);font-size:1
3.5px;}
.detail-row span:first-child{color:var(--text-muted);}
.detail-row span:last-child{font-weight:600;text-align:right;}

/* Table */
.table-wrap{overflow-x:auto;border-radius:12px;border:1px solid var(--glass-border);}
table{width:100%;border-collapse:collapse;font-size:13px;min-width:720px;}
```

```
thead th{text-align:left;padding:12px 14px;background:rgba(255,255,255,.06);color:var(--gold-light);font-size:11.5px;letter-spa
cing:.6px;text-transform:uppercase;}
tbody td{padding:11px 14px;border-top:1px solid rgba(255,255,255,.06);}
tbody tr:hover{background:rgba(255,255,255,.04);}
.credit{color:#5fe0a0;font-weight:600;}
.debit{color:#ff8a8a;font-weight:600;}
.badge{padding:3px 10px;border-radius:20px;font-size:11px;font-weight:700;}
.badge.success{background:rgba(31,168,100,.18);color:#5fe0a0;}
.filter-bar{display:flex;gap:10px;flex-wrap:wrap;margin-bottom:16px;}
.filter-bar input, .filter-bar select{padding:10px 12px;border-radius:9px;border:1px solid var(--glass-border);background:rgba
(255,255,255,.05);color:var(--text-light);font-size:13px;}

/* Toast */
.toast{position:fixed;bottom:26px;left:50%;transform:translateX(-50%) translateY(30px);background:#182338;border:1px solid var
(--gold);color:#f4f6fb;padding:14px 24px;border-radius:12px;font-size:13.5px;z-index:1500;opacity:0;transition:all .35s ease;bo
x-shadow:0 10px 30px rgba(0,0,0,.25);}
.toast.show{opacity:1;transform:translateX(-50%) translateY(0);}

.scale-in{animation:zoomIn .4s ease both;}
.hidden{display:none !important;}
::-webkit-scrollbar{width:10px;height:10px;}
::-webkit-scrollbar-track{background:transparent;}
::-webkit-scrollbar-thumb{background:rgba(255,255,255,.15);border-radius:10px;}

@media (max-width:600px){
  .auth-card{padding:28px 20px;}
  .reg-wrap{padding:34px 4% 60px;}
  .modal{padding:24px 18px;}
}
</style>
</head>
<body>
<div class="bg-shapes">
  <div class="shape s1"></div><div class="shape s2"></div><div class="shape s3"></div><div class="shape s4"></div>
</div>

<div class="app" id="app">

  <!-- ================= LANDING PAGE ================= -->
  <section class="page active" id="page-landing">
    <nav class="landing-nav">
      <div class="brand">
        <div class="logo"><div class="logo-ring"></div><div class="logo-ring r2"></div><div class="logo-core">V</div></div>
        <div class="brand-name">Vaultra <span>Bank</span></div>
      </div>
      <div class="nav-links">
        <button class="btn btn-outline btn-sm" onclick="goTo('page-login')">Login</button>
        <button class="btn btn-gold btn-sm" onclick="goTo('page-register')">Register</button>
      </div>
    </nav>

    <div class="hero">
      <div class="hero-text">
        <span class="eyebrow">Private &amp;amp; Digital Banking</span>
        <h1>Banking built like <em>a vault</em>,<br>run like <em>an app</em>.</h1>
        <p>Open a fully digital account in minutes — instant UPI, a virtual debit card, and a dashboard that shows you exactly
where your money stands, always.</p>
        <div class="hero-actions">
          <button class="btn btn-gold" onclick="goTo('page-register')">Open an Account</button>
          <button class="btn btn-outline" onclick="goTo('page-login')">Sign In</button>
        </div>
      </div>
      <div class="hero-visual">
        <div class="vault-card">
          <div class="vault-chip"></div>
          <div class="vault-num">•••• •••• •••• 4471</div>
          <div class="vault-foot"><span>VAULTRA BANK</span><span>VALID THRU 12/30</span></div>
        </div>
      </div>
    </div>

    <div class="features">
      <h2 class="section-title">Everything your money needs</h2>
      <p class="section-sub">One account, every tool, no branch visits.</p>
      <div class="feature-grid" id="featureGrid"></div>
    </div>

    <footer class="site-footer">© 2026 Vaultra Bank. A demo banking experience — no real funds are held or transferred.</footer
>
  </section>

  <!-- ================= REGISTER PAGE ================= -->
  <section class="page" id="page-register">
    <nav class="landing-nav">
      <div class="brand" style="cursor:pointer" onclick="goTo('page-landing')">
        <div class="logo"><div class="logo-ring"></div><div class="logo-ring r2"></div><div class="logo-core">V</div></div>
        <div class="brand-name">Vaultra <span>Bank</span></div>
      </div>
      <button class="btn btn-outline btn-sm" onclick="goTo('page-landing')">← Back</button>
    </nav>
```

```
    <div class="reg-wrap glass" style="padding:36px 5%;">
      <h1 class="step-title" style="font-size:26px">Open your account</h1>
      <p class="step-desc">Step <span id="stepNumLabel">1</span> of 7</p>
      <div class="progress-track"><div class="progress-fill" id="regProgressFill"></div></div>
      <div class="step-labels" id="stepLabels"></div>
      <form id="regForm" novalidate>
        <div id="stepsContainer"></div>
      </form>
    </div>
  </section>

  <!-- ================= LOGIN PAGE ================= -->
  <section class="page" id="page-login">
    <div class="auth-wrap">
      <div class="auth-card glass">
        <div class="auth-head">
          <div class="logo" style="margin:0 auto 10px;"><div class="logo-ring"></div><div class="logo-ring r2"></div><div class
="logo-core">V</div></div>
          <h2>Welcome back</h2>
          <p>Sign in to access your Vaultra account</p>
        </div>
        <form id="loginForm" novalidate>
          <div class="field" id="f_loginUsername">
            <label for="loginUsername">Username</label>
            <input type="text" id="loginUsername" placeholder="Enter your username" autocomplete="username">
            <div class="error-msg">Please enter your username</div>
          </div>
          <div class="field pw-wrap" id="f_loginPassword">
            <label for="loginPassword">Password</label>
            <input type="password" id="loginPassword" placeholder="Enter your password" autocomplete="current-password">
            <button type="button" class="pw-toggle" onclick="togglePw('loginPassword',this)">SHOW</button>
            <div class="error-msg">Please enter your password</div>
          </div>
          <div class="row-between">
            <label style="display:flex;align-items:center;gap:8px;cursor:pointer;">
              <input type="checkbox" id="rememberMe" style="accent-color:var(--gold)"> Remember me
            </label>
            <span class="link" onclick="openForgotModal()">Forgot password?</span>
          </div>
          <button type="submit" class="btn btn-gold btn-block">Login</button>
          <button type="button" class="btn btn-outline btn-block" style="margin-top:12px;" onclick="goTo('page-landing')">Back
</button>
        </form>
        <p class="auth-foot">New to Vaultra? <span class="link" onclick="goTo('page-register')">Create an account</span></p>
      </div>
    </div>
  </section>

  <!-- ================= DASHBOARD PAGE ================= -->
  <section class="page" id="page-dashboard">
    <nav class="dash-nav">
      <div class="brand">
        <div class="logo" style="width:38px;height:38px;"><div class="logo-ring"></div><div class="logo-ring r2"></div><div cla
ss="logo-core" style="font-size:12px;">V</div></div>
        <div class="brand-name" style="font-size:18px;">Vaultra <span>Bank</span></div>
      </div>
      <div class="dash-user">
        <span class="status-pill"><span class="dot"></span>ACTIVE</span>
        <div class="avatar" id="navAvatar">U</div>
        <button class="btn btn-outline btn-sm" onclick="showLogoutModal()">Logout</button>
      </div>
    </nav>
    <div class="dash-body">
      <div class="welcome-bar">
        <h1 id="welcomeName">Welcome, Customer</h1>
        <p id="welcomeSub">Here's your account at a glance</p>
      </div>

      <div class="info-grid" id="infoGrid"></div>

      <div class="section-head"><h2>Quick actions</h2></div>
      <div class="action-grid" id="actionGrid"></div>

      <div class="section-head"><h2>Recent transactions</h2><span class="link" onclick="openModal('modal-transactions')">View a
ll →</span></div>
      <div class="table-wrap">
        <table>
          <thead><tr><th>Date</th><th>Description</th><th>Credit</th><th>Debit</th><th>Balance</th></tr></thead>
          <tbody id="recentTxBody"></tbody>
        </table>
      </div>
    </div>
  </section>

</div>

<!-- ================= VERIFICATION OVERLAYS ================= -->
<div class="overlay" id="regOverlay">
  <div class="verify-box glass">
    <div id="regOverlayLoading">
```

```
      <div class="spinner"></div>
      <div class="verify-title">Setting up your account…</div>
      <div class="verify-list" id="regVerifyList"></div>
      <div class="progress-track"><div class="progress-fill" id="regOverlayFill"></div></div>
      <div class="progress-pct" id="regOverlayPct">0%</div>
    </div>
    <div id="regOverlaySuccess" class="hidden">
      <div class="success-check"><div class="circle"><svg viewBox="0 0 24 24" fill="none"><path d="M5 13l4 4L19 7" stroke="#ff
f" stroke-width="3" stroke-linecap="round" stroke-linejoin="round"/></svg></div></div>
      <div class="success-title">🎉 Registration Successful!</div>
      <p class="success-sub">Redirecting you to login…</p>
    </div>
  </div>
</div>

<div class="overlay" id="loginOverlay">
  <div class="verify-box glass">
    <div id="loginOverlayLoading">
      <div class="spinner"></div>
      <div class="verify-title">Verifying Credentials…</div>
      <div class="progress-track"><div class="progress-fill" id="loginOverlayFill"></div></div>
      <div class="progress-pct" id="loginOverlayPct">0%</div>
    </div>
    <div id="loginOverlaySuccess" class="hidden">
      <div class="success-check"><div class="circle"><svg viewBox="0 0 24 24" fill="none"><path d="M5 13l4 4L19 7" stroke="#ff
f" stroke-width="3" stroke-linecap="round" stroke-linejoin="round"/></svg></div></div>
      <div class="success-title">Authentication Successful</div>
      <p class="success-sub">Welcome back!</p>
    </div>
  </div>
</div>

<!-- ================= MODALS ================= -->
<div class="modal-overlay" id="modalRoot"></div>

<div class="toast" id="toast"></div>

<script>
/* ========================================================================
   VAULTRA BANK — application script
======================================================================== */

/* ---------- Utility: Ripple effect on all .btn ---------- */
document.addEventListener('click', function(e){
  const btn = e.target.closest('.btn, .action-card, .toggle-item, .feature-card');
  if(!btn) return;
  const rect = btn.getBoundingClientRect();
  const ripple = document.createElement('span');
  ripple.className='ripple';
  const size = Math.max(rect.width, rect.height);
  ripple.style.width = ripple.style.height = size+'px';
  ripple.style.left = (e.clientX - rect.left - size/2)+'px';
  ripple.style.top = (e.clientY - rect.top - size/2)+'px';
  btn.style.position = btn.style.position || 'relative';
  btn.style.overflow = 'hidden';
  btn.appendChild(ripple);
  setTimeout(()=>ripple.remove(), 650);
});

/* ---------- Toast ---------- */
let toastTimer;
function showToast(msg){
  const t = document.getElementById('toast');
  t.textContent = msg;
  t.classList.add('show');
  clearTimeout(toastTimer);
  toastTimer = setTimeout(()=>t.classList.remove('show'), 2800);
}

/* ---------- Page navigation ---------- */
function goTo(pageId){
  const current = document.querySelector('.page.active');
  const next = document.getElementById(pageId);
  if(current === next) return;
  if(current){
    current.classList.add('leaving');
    setTimeout(()=>{
      current.classList.remove('active','leaving');
      next.classList.add('active');
      window.scrollTo({top:0,behavior:'smooth'});
    }, 320);
  } else {
    next.classList.add('active');
  }
}

/* ---------- Password show/hide ---------- */
function togglePw(id, btn){
  const el = document.getElementById(id);
  if(el.type === 'password'){ el.type='text'; btn.textContent='HIDE'; }
```

```
  else { el.type='password'; btn.textContent='SHOW'; }
}

/* ---------- Storage helpers ---------- */
const DB = {
  getUsers(){ return JSON.parse(localStorage.getItem('vaultra_users') || '{}'); },
  saveUsers(u){ localStorage.setItem('vaultra_users', JSON.stringify(u)); },
  getUser(username){ return this.getUsers()[username]; },
  saveUser(user){ const u = this.getUsers(); u[user.security.username] = user; this.saveUsers(u); },
  setCurrent(username){ localStorage.setItem('vaultra_current', username); },
  getCurrent(){ return localStorage.getItem('vaultra_current'); },
  clearCurrent(){ localStorage.removeItem('vaultra_current'); }
};

/* ---------- Modal system ---------- */
function openModal(html, isId){
  const root = document.getElementById('modalRoot');
  if(isId === undefined && document.getElementById(html)){
    // pre-registered template id (like modal-logout) not used directly; fallback
  }
  root.innerHTML = html;
  root.classList.add('active');
}
function closeModal(){
  document.getElementById('modalRoot').classList.remove('active');
}
function modalShell(title, bodyHtml, extraClass){
  return `<div class="modal glass ${extraClass||''}">
    <button class="modal-close" onclick="closeModal()">×</button>
    <h2>${title}</h2>
    ${bodyHtml}
  </div>`;
}

/* ---------- Confetti ---------- */
function launchConfetti(){
  const colors = ['#c9a227','#e8c766','#3b5fd9','#ffffff','#5fe0a0'];
  for(let i=0;i<70;i++){
    const p = document.createElement('div');
    p.className='confetti-piece';
    p.style.left = Math.random()*100+'vw';
    p.style.background = colors[Math.floor(Math.random()*colors.length)];
    p.style.animationDuration = (2.2+Math.random()*1.6)+'s';
    p.style.transform = `rotate(${Math.random()*360}deg)`;
    document.body.appendChild(p);
    setTimeout(()=>p.remove(), 4200);
  }
}

/* ---------- Random helpers for realistic banking data ---------- */
function randDigits(n){ let s=''; for(let i=0;i<n;i++) s+=Math.floor(Math.random()*10); return s; }
function pad(n,len){ return String(n).padStart(len,'0'); }
function fmtMoney(v){ return '₹' + Number(v).toLocaleString('en-IN', {minimumFractionDigits:2, maximumFractionDigits:2}); }
function fmtDate(d){ const dt = new Date(d); return dt.toLocaleDateString('en-IN', {day:'2-digit',month:'short',year:'numeri
c'}); }
function fmtDateTime(d){ const dt = new Date(d); return dt.toLocaleString('en-IN', {day:'2-digit',month:'short',year:'numeric',
hour:'2-digit',minute:'2-digit'}); }

const BRANCHES = {
  "Connaught Place, New Delhi": "VLTB0001101",
  "Bandra Kurla Complex, Mumbai": "VLTB0001102",
  "MG Road, Bengaluru": "VLTB0001103",
  "Salt Lake, Kolkata": "VLTB0001104",
  "Anna Salai, Chennai": "VLTB0001105",
  "Banjara Hills, Hyderabad": "VLTB0001106",
  "Koregaon Park, Pune": "VLTB0001107",
  "Sector 17, Chandigarh": "VLTB0001108"
};

/* ======================================================
   LANDING FEATURES
====================================================== */
const FEATURES = [
  {icon:'⚡', title:'Instant Account Opening', desc:'Go from application to active account in under 5 minutes, fully online.'},
  {icon:'💳', title:'Free Virtual Debit Card', desc:'Get an instant virtual debit card the moment your account is created.'},
  {icon:'📲', title:'UPI & Mobile Banking', desc:'Pay, request, and track spending from one clean, fast mobile dashboard.'},
  {icon:'🔒', title:'Bank-Grade Security', desc:'Transaction PINs, security questions, and encrypted local storage.'},
  {icon:'📈', title:'Fixed Deposits', desc:'Lock in guaranteed returns with flexible tenures and instant calculators.'},
  {icon:'🏦', title:'8 Branches Nationwide', desc:'Choose a home branch from major cities across India at signup.'}
];
function renderFeatures(){
  const grid = document.getElementById('featureGrid');
  grid.innerHTML = FEATURES.map((f,i)=>`
    <div class="feature-card glass" style="animation-delay:${i*0.08}s">
      <div class="feature-icon">${f.icon}</div>
      <h3>${f.title}</h3>
      <p>${f.desc}</p>
    </div>`).join('');
}
renderFeatures();
```

```
/* ====================================================
   REGISTRATION — field configuration (7 steps)
==================================================== */
const STEP_META = [
  {key:'personal', label:'Personal', title:'Personal Information', desc:"Tell us who you are — exactly as it appears on your ID documents."},
  {key:'contact', label:'Contact', title:'Contact Information', desc:'How can we reach you, and where do you live?'},
  {key:'employment', label:'Employment', title:'Employment Details', desc:'Help us understand your income and occupation.'},
  {key:'banking', label:'Banking', title:'Banking Details', desc:'Choose your branch, account type and services.'},
  {key:'security', label:'Security', title:'Account Security', desc:'Create your login credentials and transaction PIN.'},
  {key:'nominee', label:'Nominee', title:'Nominee Details', desc:'Add a nominee to your account for added protection.'},
  {key:'documents', label:'Documents', title:'Upload Documents', desc:'Upload the documents required to verify your identity.'}
];

const FIELDS = {
  personal: [
    {id:'fullName', label:"Full Name", type:'text', req:true, ph:'As per Aadhaar / PAN', pattern:'name'},
    {id:'fatherName', label:"Father's Name", type:'text', req:true, pattern:'name'},
    {id:'motherName', label:"Mother's Name", type:'text', req:true, pattern:'name'},
    {id:'dob', label:'Date of Birth', type:'date', req:true},
    {id:'gender', label:'Gender', type:'select', req:true, opts:['Male','Female','Other']},
    {id:'maritalStatus', label:'Marital Status', type:'select', req:true, opts:['Single','Married','Divorced','Widowed']},
    {id:'nationality', label:'Nationality', type:'text', req:true, val:'Indian'},
    {id:'bloodGroup', label:'Blood Group', type:'select', req:true, opts:['A+','A-','B+','B-','O+','O-','AB+','AB-']},
    {id:'aadhaar', label:'Aadhaar Number', type:'text', req:true, ph:'12-digit number', maxlen:12, pattern:'aadhaar'},
    {id:'pan', label:'PAN Number', type:'text', req:true, ph:'ABCDE1234F', maxlen:10, pattern:'pan'},
    {id:'passport', label:'Passport Number', type:'text', req:false, ph:'Optional'},
    {id:'drivingLicense', label:'Driving License Number', type:'text', req:false, ph:'Optional'},
    {id:'voterId', label:'Voter ID', type:'text', req:false, ph:'Optional'}
  ],
  contact: [
    {id:'mobile', label:'Mobile Number', type:'text', req:true, ph:'10-digit mobile number', maxlen:10, pattern:'phone'},
    {id:'altMobile', label:'Alternate Mobile Number', type:'text', req:false, maxlen:10, pattern:'phone'},
    {id:'email', label:'Email', type:'email', req:true, ph:'you@example.com', pattern:'email'},
    {id:'altEmail', label:'Alternate Email', type:'email', req:false},
    {id:'houseNo', label:'House Number', type:'text', req:true},
    {id:'street', label:'Street', type:'text', req:true},
    {id:'area', label:'Area', type:'text', req:true},
    {id:'landmark', label:'Landmark', type:'text', req:false},
    {id:'city', label:'City', type:'text', req:true},
    {id:'district', label:'District', type:'text', req:true},
    {id:'state', label:'State', type:'text', req:true},
    {id:'country', label:'Country', type:'text', req:true, val:'India'},
    {id:'pincode', label:'PIN Code', type:'text', req:true, maxlen:6, pattern:'pincode'}
  ],
  employment: [
    {id:'occupation', label:'Occupation', type:'text', req:true},
    {id:'employmentType', label:'Employment Type', type:'select', req:true, opts:['Salaried','Self-Employed','Business Owner','Student','Retired','Unemployed']},
    {id:'companyName', label:'Company Name', type:'text', req:false},
    {id:'designation', label:'Designation', type:'text', req:false},
    {id:'workExperience', label:'Work Experience (years)', type:'number', req:false, pattern:'number'},
    {id:'monthlyIncome', label:'Monthly Income (₹)', type:'number', req:true, pattern:'number'},
    {id:'annualIncome', label:'Annual Income (₹)', type:'number', req:true, pattern:'number'},
    {id:'education', label:'Educational Qualification', type:'select', req:true, opts:['High School','Diploma',"Bachelor's Degree","Master's Degree",'Doctorate','Other']}
  ],
  banking: [
    {id:'branch', label:'Branch Selection', type:'select', req:true, opts:Object.keys(BRANCHES)},
    {id:'accountType', label:'Account Type', type:'select', req:true, opts:['Savings Account','Current Account','Salary Account','Senior Citizen Account']},
    {id:'initialDeposit', label:'Initial Deposit (₹)', type:'number', req:true, ph:'Minimum ₹500'}
  ],
  bankingToggles: [
    {id:'debitCard', label:'Debit Card Required'},
    {id:'creditCard', label:'Credit Card Required'},
    {id:'chequeBook', label:'Cheque Book'},
    {id:'passbook', label:'Passbook'},
    {id:'internetBanking', label:'Internet Banking'},
    {id:'mobileBanking', label:'Mobile Banking'},
    {id:'smsAlerts', label:'SMS Alerts'},
    {id:'upiRegistration', label:'UPI Registration'},
    {id:'atmFacility', label:'ATM Facility'}
  ],
  security: [
    {id:'username', label:'Username', type:'text', req:true, ph:'Choose a unique username'},
    {id:'password', label:'Password', type:'password', req:true, ph:'Min 8 chars, 1 upper, 1 number, 1 symbol', pattern:'password'},
    {id:'confirmPassword', label:'Confirm Password', type:'password', req:true, pattern:'confirmPassword'},
    {id:'transactionPin', label:'Transaction PIN', type:'text', req:true, ph:'4-digit PIN', maxlen:4, pattern:'pin'},
    {id:'securityQuestion', label:'Security Question', type:'select', req:true, opts:["What is your pet's name?","What city were you born in?","What is your favorite teacher's name?","What was your first school's name?"]},
    {id:'securityAnswer', label:'Security Answer', type:'text', req:true}
  ],
  nominee: [
    {id:'nomineeName', label:'Nominee Name', type:'text', req:true, pattern:'name'},
    {id:'relationship', label:'Relationship', type:'select', req:true, opts:['Spouse','Son','Daughter','Father','Mother','Sibling','Other']},
    {id:'nomineeAge', label:'Age', type:'number', req:true, pattern:'number'},
```

```
    {id:'nomineeMobile', label:'Mobile Number', type:'text', req:true, maxlen:10, pattern:'phone'},
    {id:'nomineeAddress', label:'Address', type:'textarea', req:true}
  ],
  documents: [
    {id:'passportPhoto', label:'Passport Photo'},
    {id:'aadhaarCard', label:'Aadhaar Card'},
    {id:'panCard', label:'PAN Card'},
    {id:'signature', label:'Signature'},
    {id:'addressProof', label:'Address Proof'}
  ]
};

let regState = { step:1, data:{}, uploads:{}, avatar:null };
const TOTAL_STEPS = 7;

/* ====================================================
   REGISTRATION — render steps
==================================================== */
const RESTRICT_MAP = { name:'letters', phone:'digits', aadhaar:'digits', pincode:'digits', pin:'digits' };
function restrictInput(el, kind){
  if(kind === 'digits') el.value = el.value.replace(/[^0-9]/g,'');
  else if(kind === 'letters') el.value = el.value.replace(/[^A-Za-z\s'.-]/g,'');
}
function fieldHtml(f, stepKey){
  const errId = `err_${f.id}`;
  const restrict = RESTRICT_MAP[f.pattern];
  const restrictAttr = restrict ? ` oninput="restrictInput(this,'${restrict}')" inputmode="${restrict==='digits'?'numeric':'tex
t'}"` : '';
  let control = '';
  if(f.type === 'select'){
    control = `<select id="${f.id}">
      <option value="">Select ${f.label}</option>
      ${f.opts.map(o=>`<option value="${o}">${o}</option>`).join('')}
    </select>`;
  } else if(f.type === 'textarea'){
    control = `<textarea id="${f.id}" rows="3" placeholder="${f.ph||''}"></textarea>`;
  } else if(f.type === 'password'){
    control = `<div class="pw-wrap"><input type="password" id="${f.id}" placeholder="${f.ph||''}">
      <button type="button" class="pw-toggle" onclick="togglePw('${f.id}',this)">SHOW</button></div>`;
  } else {
    control = `<input type="${f.type}" id="${f.id}" placeholder="${f.ph||''}" ${f.maxlen?`maxlength="${f.maxlen}"`:''} value
="${f.val||''}"${restrictAttr}>`;
  }
  return `<div class="field" id="f_${f.id}">
    <label for="${f.id}">${f.label}${f.req?' *':''}</label>
    ${control}
    <div class="error-msg" id="${errId}">This field is required</div>
  </div>`;
}

function renderStepLabels(){
  const el = document.getElementById('stepLabels');
  el.innerHTML = STEP_META.map((s,i)=>`<span id="lbl_${i+1}" class="${i+1===regState.step?'active':''}">${s.label}</span>`).joi
n('');
}

function renderSteps(){
  const container = document.getElementById('stepsContainer');
  let html = '';
  STEP_META.forEach((meta, idx)=>{
    const stepNum = idx+1;
    html += `<div class="form-step ${stepNum===regState.step?'active':''}" data-step="${stepNum}">
      <div class="step-title">${meta.title}</div>
      <div class="step-desc">${meta.desc}</div>`;

    if(meta.key === 'banking'){
      html += `<div class="grid2">${FIELDS.banking.map(f=>fieldHtml(f)).join('')}</div>`;
      html += `<label style="display:block;font-size:13px;color:var(--text-muted);margin:10px 0 10px;font-weight:600;">Select t
he services you'd like activated:</label>`;
      html += `<div class="check-grid">${FIELDS.bankingToggles.map(t=>`
        <div class="toggle-item"><span>${t.label}</span>
          <label class="switch"><input type="checkbox" id="${t.id}" checked><span class="slider-toggle"></span></label>
        </div>`).join('')}</div>`;
    } else if(meta.key === 'documents'){
      html += FIELDS.documents.map(d=>`
        <div class="upload-box" id="box_${d.id}">
          <input type="file" id="${d.id}" accept="image/*,.pdf" onchange="handleUpload('${d.id}')">
          <div>📎 <strong>${d.label}</strong> — click or drop file to upload</div>
          <div class="fname" id="fname_${d.id}"></div>
        </div>`).join('');
      html += `<div class="check-row"><input type="checkbox" id="agreeTerms"> <span>I have read and agree to the <span class="l
ink">Terms &amp;amp; Conditions</span> *</span></div>
        <div class="error-msg" id="err_agreeTerms">You must accept the Terms &amp;amp; Conditions</div>
        <div class="check-row"><input type="checkbox" id="agreePrivacy"> <span>I have read and agree to the <span class="link">
Privacy Policy</span> *</span></div>
        <div class="error-msg" id="err_agreePrivacy">You must accept the Privacy Policy</div>
        <div class="check-row"><input type="checkbox" id="agreeDeclaration"> <span>I hereby declare that the information provid
ed is true and correct to the best of my knowledge *</span></div>
        <div class="error-msg" id="err_agreeDeclaration">You must accept the declaration</div>`;
    } else {
```

```
      html += `<div class="grid2">${FIELDS[meta.key].map(f=>fieldHtml(f)).join('')}</div>`;
    }

    html += `<div class="step-nav">
      ${stepNum>1 ? `<button type="button" class="btn btn-outline" onclick="prevStep()">← Back</button>` : `<button type="butto
n" class="btn btn-outline" onclick="goTo('page-landing')">Cancel</button>`}
      ${stepNum<TOTAL_STEPS ? `<button type="button" class="btn btn-royal" onclick="nextStep()">Continue →</button>` : `<button
type="button" class="btn btn-gold" onclick="finishRegistration()">Finish Registration ✓</button>`}
    </div></div>`;
  });
  container.innerHTML = html;
  restoreStepValues();
}

function handleUpload(id){
  const input = document.getElementById(id);
  const box = document.getElementById('box_'+id);
  const label = document.getElementById('fname_'+id);
  if(input.files && input.files[0]){
    const file = input.files[0];
    label.textContent = '✓ ' + file.name;
    box.style.borderColor = 'var(--success)';
    regState.uploads[id] = file.name;
    if(id === 'passportPhoto'){
      const reader = new FileReader();
      reader.onload = e => { regState.avatar = e.target.result; };
      reader.readAsDataURL(file);
    }
  }
}

function restoreStepValues(){
  Object.keys(regState.data).forEach(k=>{
    const el = document.getElementById(k);
    if(el && el.type !== 'file') el.value = regState.data[k];
  });
  FIELDS.bankingToggles.forEach(t=>{
    const el = document.getElementById(t.id);
    if(el && regState.data[t.id] !== undefined) el.checked = regState.data[t.id];
  });
  ['agreeTerms','agreePrivacy','agreeDeclaration'].forEach(id=>{
    const el = document.getElementById(id);
    if(el && regState.data[id]) el.checked = true;
  });
}

function updateProgress(){
  const pct = Math.round((regState.step-1)/(TOTAL_STEPS-1)*100);
  document.getElementById('regProgressFill').style.width = pct+'%';
  document.getElementById('stepNumLabel').textContent = regState.step;
  STEP_META.forEach((s,i)=>{
    const lbl = document.getElementById('lbl_'+(i+1));
    if(!lbl) return;
    lbl.classList.toggle('active', i+1===regState.step);
    lbl.classList.toggle('done', i+1<regState.step);
  });
}

/* ---------- Validation patterns ---------- */
const PATTERNS = {
  email: /^[^\s@]+@[^\s@]+\.[^\s@]+$/,
  phone: /^[6-9]\d{9}$/,
  aadhaar: /^\d{12}$/,
  pan: /^[A-Z]{5}\d{4}[A-Z]$/,
  pincode: /^\d{6}$/,
  pin: /^\d{4,6}$/,
  password: /^(?=.*[A-Z])(?=.*\d)(?=.*[!@#$%^&*()_+\-=]).{8,}$/,
  name: /^[A-Za-z][A-Za-z\s.'-]{1,49}$/,
  number: /^\d+$/
};

function setFieldError(id, hasError, msg){
  const wrap = document.getElementById('f_'+id) || document.getElementById('box_'+id) || (document.getElementById(id) ? documen
t.getElementById(id).closest('.check-row') : null);
  const err = document.getElementById('err_'+id);
  if(wrap) wrap.classList.toggle('error', hasError);
  if(err && msg) err.textContent = msg;
}

function validateStep(stepNum){
  let valid = true;
  const meta = STEP_META[stepNum-1];

  if(meta.key === 'banking'){
    FIELDS.banking.forEach(f=>{
      const el = document.getElementById(f.id);
      const val = el.value.trim();
      regState.data[f.id] = val;
      let err = false, msg='This field is required';
      if(f.req && !val) err = true;
```

```
      else if(f.id === 'initialDeposit' && val && Number(val) < 500){ err = true; msg='Minimum initial deposit is ₹500'; }
      setFieldError(f.id, err, msg);
      if(err) valid = false;
    });
    FIELDS.bankingToggles.forEach(t=>{ regState.data[t.id] = document.getElementById(t.id).checked; });
    return valid;
  }

  if(meta.key === 'documents'){
    FIELDS.documents.forEach(d=>{
      if(!regState.uploads[d.id]){ setFieldError(d.id, true); valid = false; }
      else setFieldError(d.id, false);
    });
    ['agreeTerms','agreePrivacy','agreeDeclaration'].forEach(id=>{
      const checked = document.getElementById(id).checked;
      regState.data[id] = checked;
      setFieldError(id, !checked);
      if(!checked) valid = false;
    });
    return valid;
  }

  const fields = FIELDS[meta.key];
  fields.forEach(f=>{
    const el = document.getElementById(f.id);
    const val = el.value.trim();
    regState.data[f.id] = val;
    let err = false, msg = 'This field is required';
    if(f.req && !val){ err = true; }
    else if(val && f.pattern){
      if(f.pattern === 'email' && !PATTERNS.email.test(val)){ err=true; msg='Enter a valid email address'; }
      else if(f.pattern === 'phone' && !PATTERNS.phone.test(val)){ err=true; msg='Enter a valid 10-digit mobile number'; }
      else if(f.pattern === 'aadhaar' && !PATTERNS.aadhaar.test(val)){ err=true; msg='Aadhaar must be exactly 12 digits'; }
      else if(f.pattern === 'pan' && !PATTERNS.pan.test(val.toUpperCase())){ err=true; msg='PAN format: ABCDE1234F'; }
      else if(f.pattern === 'pincode' && !PATTERNS.pincode.test(val)){ err=true; msg='PIN code must be 6 digits'; }
      else if(f.pattern === 'pin' && !PATTERNS.pin.test(val)){ err=true; msg='PIN must be 4-6 digits'; }
      else if(f.pattern === 'password' && !PATTERNS.password.test(val)){ err=true; msg='Min 8 chars with 1 uppercase, 1 number
& 1 symbol'; }
      else if(f.pattern === 'confirmPassword' && val !== document.getElementById('password').value){ err=true; msg='Passwords d
o not match'; }
      else if(f.pattern === 'name' && !PATTERNS.name.test(val)){ err=true; msg='Enter a valid name (letters only, 2-50 characte
rs)'; }
      else if(f.pattern === 'number' && (!PATTERNS.number.test(val) || Number(val) <= 0)){ err=true; msg='Enter a valid number
greater than 0'; }
    }
    setFieldError(f.id, err, msg);
    if(err) valid = false;
  });
  return valid;
}

function nextStep(){
  if(!validateStep(regState.step)){ showToast('Please fix the highlighted fields'); return; }
  if(regState.step < TOTAL_STEPS){
    regState.step++;
    document.querySelectorAll('.form-step').forEach(s=>s.classList.remove('active'));
    document.querySelector(`.form-step[data-step="${regState.step}"]`).classList.add('active');
    updateProgress();
    restoreStepValues();
    document.querySelector('.reg-wrap').scrollIntoView({behavior:'smooth'});
  }
}
function prevStep(){
  if(regState.step > 1){
    regState.step--;
    document.querySelectorAll('.form-step').forEach(s=>s.classList.remove('active'));
    document.querySelector(`.form-step[data-step="${regState.step}"]`).classList.add('active');
    updateProgress();
  }
}

/* ====================================================
   REGISTRATION — finish & account creation
==================================================== */
const VERIFY_STEPS = [
  'Verifying Personal Details',
  'Checking Identity Documents',
  'Creating Customer ID',
  'Generating Account Number',
  'Creating Debit Card',
  'Activating Internet Banking',
  'Creating UPI ID',
  'Finalizing Registration'
];

function finishRegistration(){
  if(!validateStep(TOTAL_STEPS)){ showToast('Please complete all required fields'); return; }

  if(DB.getUser(regState.data.username)){
    showToast('Username already taken — please choose another');
```

```
    regState.step = 5;
    document.querySelectorAll('.form-step').forEach(s=>s.classList.remove('active'));
    document.querySelector(`.form-step[data-step="5"]`).classList.add('active');
    updateProgress();
    return;
  }

  const overlay = document.getElementById('regOverlay');
  const listEl = document.getElementById('regVerifyList');
  listEl.innerHTML = VERIFY_STEPS.map((s,i)=>`<div class="verify-item" id="rvi_${i}"><span class="tick"></span><span>${s}</span
></div>`).join('');
  document.getElementById('regOverlayLoading').classList.remove('hidden');
  document.getElementById('regOverlaySuccess').classList.add('hidden');
  overlay.classList.add('active');

  const fill = document.getElementById('regOverlayFill');
  const pct = document.getElementById('regOverlayPct');
  let i = 0;
  fill.style.width = '0%'; pct.textContent = '0%';

  const timer = setInterval(()=>{
    if(i < VERIFY_STEPS.length){
      document.getElementById('rvi_'+i).classList.add('done');
      i++;
      const p = Math.round((i/VERIFY_STEPS.length)*100);
      fill.style.width = p+'%';
      pct.textContent = p+'%';
    } else {
      clearInterval(timer);
      createAccount();
      setTimeout(()=>{
        document.getElementById('regOverlayLoading').classList.add('hidden');
        document.getElementById('regOverlaySuccess').classList.remove('hidden');
        launchConfetti();
        setTimeout(()=>{
          overlay.classList.remove('active');
          goTo('page-login');
          document.getElementById('loginUsername').value = regState.data.username;
          showToast('Account created! Please sign in.');
        }, 3000);
      }, 400);
    }
  }, 420);
}

function createAccount(){
  const d = regState.data;
  const now = new Date().toISOString();
  const customerId = 'CUST' + randDigits(8);
  const accountNumber = randDigits(16);
  const cif = 'CIF' + randDigits(9);
  const ifsc = BRANCHES[d.branch] || 'VLTB0001100';
  const upiId = d.username + '@vaultra';

  const user = {
    personal: {}, contact: {}, employment: {}, banking: {}, nominee: {}, documents: regState.uploads, security: {},
    avatar: regState.avatar
  };
  FIELDS.personal.forEach(f=>user.personal[f.id]=d[f.id]||'');
  FIELDS.contact.forEach(f=>user.contact[f.id]=d[f.id]||'');
  FIELDS.employment.forEach(f=>user.employment[f.id]=d[f.id]||'');
  FIELDS.banking.forEach(f=>user.banking[f.id]=d[f.id]||'');
  FIELDS.bankingToggles.forEach(t=>user.banking[t.id]=!!d[t.id]);
  FIELDS.nominee.forEach(f=>user.nominee[f.id]=d[f.id]||'');
  FIELDS.security.forEach(f=>{ if(f.id!=='confirmPassword') user.security[f.id]=d[f.id]||''; });

  user.account = {
    customerId, accountNumber, cif, ifsc,
    branch: d.branch,
    accountType: d.accountType,
    balance: Number(d.initialDeposit || 500),
    ledgerBalance: Number(d.initialDeposit || 500),
    upiId,
    kycStatus: 'Verified',
    accountStatus: 'ACTIVE',
    openingDate: now,
    lastLogin: null,
    cardExpiry: (function(){ const dt=new Date(); dt.setFullYear(dt.getFullYear()+5); return pad(dt.getMonth()+1,2)+'/'+String
(dt.getFullYear()).slice(2); })(),
    cardNumber: randDigits(4)+' '+randDigits(4)+' '+randDigits(4)+' '+randDigits(4)
  };

  // Seed realistic transaction history
  user.transactions = generateSeedTransactions(user.account.balance);

  DB.saveUser(user);
}

function generateSeedTransactions(finalBalance){
  const descs = [
```

```
    {d:'Initial Deposit', type:'credit'},
    {d:'ATM Withdrawal - Vaultra ATM', type:'debit'},
    {d:'UPI - Swiggy Food Delivery', type:'debit'},
    {d:'Salary Credit', type:'credit'},
    {d:'UPI - Electricity Bill', type:'debit'},
    {d:'Interest Credit', type:'credit'},
    {d:'UPI Transfer to Friend', type:'debit'},
    {d:'Online Shopping - Amazon', type:'debit'},
    {d:'Mobile Recharge', type:'debit'},
    {d:'Refund - Cancelled Order', type:'credit'}
  ];
  let balance = 500;
  const txns = [];
  const startDate = new Date();
  startDate.setDate(startDate.getDate()-30);
  descs.forEach((item,idx)=>{
    const amt = item.type==='credit' ? Math.round(500+Math.random()*15000) : Math.round(100+Math.random()*4000);
    if(item.type==='credit') balance += amt; else balance = Math.max(0, balance-amt);
    const date = new Date(startDate.getTime() + idx*(2.6*24*3600*1000));
    txns.push({
      id: 'TXN'+randDigits(10),
      date: date.toISOString(),
      description: item.d,
      credit: item.type==='credit'?amt:0,
      debit: item.type==='debit'?amt:0,
      balance: balance,
      status: 'Success'
    });
  });
  // adjust so it ends near current known balance, then add final deposit-in-line entry
  const diff = finalBalance - balance;
  balance += diff;
  txns.push({
    id:'TXN'+randDigits(10), date:new Date().toISOString(), description:'Account Opening Deposit',
    credit: diff>0?diff:0, debit: diff<0?Math.abs(diff):0, balance: finalBalance, status:'Success'
  });
  return txns;
}

/* ====================================================
   LOGIN
==================================================== */
document.getElementById('loginForm').addEventListener('submit', function(e){
  e.preventDefault();
  const username = document.getElementById('loginUsername').value.trim();
  const password = document.getElementById('loginPassword').value;
  let valid = true;
  setFieldError('loginUsername', !username); if(!username) valid=false;
  setFieldError('loginPassword', !password); if(!password) valid=false;
  if(!valid){ showToast('Please enter your credentials'); return; }

  const user = DB.getUser(username);
  if(!user || user.security.password !== password){
    setFieldError('loginUsername', true, 'Invalid username or password');
    setFieldError('loginPassword', true, 'Invalid username or password');
    showToast('Invalid username or password');
    return;
  }

  const overlay = document.getElementById('loginOverlay');
  document.getElementById('loginOverlayLoading').classList.remove('hidden');
  document.getElementById('loginOverlaySuccess').classList.add('hidden');
  overlay.classList.add('active');
  const fill = document.getElementById('loginOverlayFill');
  const pct = document.getElementById('loginOverlayPct');
  fill.style.width='0%'; pct.textContent='0%';
  let p = 0;
  const timer = setInterval(()=>{
    p += 12 + Math.random()*10;
    if(p >= 100){
      p = 100;
      clearInterval(timer);
      fill.style.width='100%'; pct.textContent='100%';
      setTimeout(()=>{
        document.getElementById('loginOverlayLoading').classList.add('hidden');
        document.getElementById('loginOverlaySuccess').classList.remove('hidden');
        launchConfetti();
        user.account.lastLoginPrevious = user.account.lastLogin;
        user.account.lastLogin = new Date().toISOString();
        DB.saveUser(user);
        DB.setCurrent(username);
        setTimeout(()=>{
          overlay.classList.remove('active');
          loadDashboard(username);
          goTo('page-dashboard');
        }, 1400);
      }, 350);
    } else {
      fill.style.width = p+'%'; pct.textContent = Math.round(p)+'%';
    }
```

```
  }, 200);
});

function openForgotModal(){
  openModal(modalShell('Forgot Password', `
    <p style="color:var(--text-muted);font-size:13.5px;margin-bottom:18px;">Enter your registered username and we'll verify you
r security question to help you reset access.</p>
    <div class="field"><label>Username</label><input type="text" id="fpUsername" placeholder="Your username"></div>
    <div id="fpQuestionArea"></div>
    <button class="btn btn-gold btn-block" onclick="fpCheckUser()">Continue</button>
  `));
}
function fpCheckUser(){
  const uname = document.getElementById('fpUsername').value.trim();
  const user = DB.getUser(uname);
  const area = document.getElementById('fpQuestionArea');
  if(!user){ showToast('No account found with that username'); return; }
  area.innerHTML = `
    <div class="field"><label>${user.security.securityQuestion}</label><input type="text" id="fpAnswer" placeholder="Your answe
r"></div>
    <div class="field pw-wrap"><label>New Password</label><input type="password" id="fpNewPw" placeholder="Min 8 chars, 1 uppe
r, 1 number, 1 symbol"></div>
    <button class="btn btn-royal btn-block" onclick="fpReset('${uname}')">Reset Password</button>`;
}
function fpReset(uname){
  const user = DB.getUser(uname);
  const ans = document.getElementById('fpAnswer').value.trim();
  const newPw = document.getElementById('fpNewPw').value;
  if(ans.toLowerCase() !== (user.security.securityAnswer||'').toLowerCase()){ showToast('Security answer incorrect'); return; }
  if(!PATTERNS.password.test(newPw)){ showToast('Password does not meet strength requirements'); return; }
  user.security.password = newPw;
  DB.saveUser(user);
  showToast('Password reset! Please log in.');
  closeModal();
}

/* ======================================================
   DASHBOARD
====================================================== */
let currentUser = null;

const ACTIONS = [
  {id:'view-profile', icon:'👤', label:'View Profile'},
  {id:'edit-profile', icon:'✏️', label:'Edit Profile'},
  {id:'deposit', icon:'⬇️', label:'Deposit Money'},
  {id:'withdraw', icon:'⬆️', label:'Withdraw Money'},
  {id:'transfer', icon:'🔁', label:'Transfer Money'},
  {id:'transactions', icon:'📜', label:'Transaction History'},
  {id:'passbook', icon:'📥', label:'Download Passbook'},
  {id:'card', icon:'💳', label:'Card Details'},
  {id:'loan', icon:'🏠', label:'Loan Eligibility'},
  {id:'fd', icon:'📈', label:'Fixed Deposit'},
  {id:'settings', icon:'⚙️', label:'Settings'},
  {id:'logout', icon:'🚪', label:'Logout'}
];

function loadDashboard(username){
  currentUser = DB.getUser(username);
  if(!currentUser) return;
  renderDashboard();
}

function renderDashboard(){
  const u = currentUser;
  document.getElementById('welcomeName').textContent = 'Welcome, ' + (u.personal.fullName || 'Customer');
  document.getElementById('welcomeSub').textContent = `Customer ID ${u.account.customerId} · Last login ${u.account.lastLoginPr
evious ? fmtDateTime(u.account.lastLoginPrevious) : 'First login'}`;

  const initials = (u.personal.fullName||'U').trim().split(' ').map(s=>s[0]).slice(0,2).join('').toUpperCase();
  const avatarEls = [document.getElementById('navAvatar')];
  avatarEls.forEach(el=>{
    el.innerHTML = u.avatar ? `<img src="${u.avatar}">` : initials;
  });

  document.getElementById('infoGrid').innerHTML = `
    <div class="info-card glass balance"><div class="label">Available Balance</div><div class="value">${fmtMoney(u.account.bala
nce)}</div></div>
    <div class="info-card glass"><div class="label">Ledger Balance</div><div class="value">${fmtMoney(u.account.ledgerBalance)}
</div></div>
    <div class="info-card glass"><div class="label">Account Number</div><div class="value">${u.account.accountNumber}</div></di
v>
    <div class="info-card glass"><div class="label">Customer ID</div><div class="value">${u.account.customerId}</div></div>
    <div class="info-card glass"><div class="label">CIF Number</div><div class="value">${u.account.cif}</div></div>
    <div class="info-card glass"><div class="label">IFSC Code</div><div class="value">${u.account.ifsc}</div></div>
    <div class="info-card glass"><div class="label">Branch</div><div class="value" style="font-size:14px;">${u.account.branch}
</div></div>
    <div class="info-card glass"><div class="label">Account Type</div><div class="value" style="font-size:15px;">${u.account.ac
countType}</div></div>
    <div class="info-card glass"><div class="label">UPI ID</div><div class="value" style="font-size:15px;">${u.account.upiId}</
div></div>
```

```
    <div class="info-card glass"><div class="label">Internet Banking</div><div class="value" style="font-size:15px;color:${u.ba
nking.internetBanking?'#5fe0a0':'#ff8a8a'}">${u.banking.internetBanking?'Active':'Inactive'}</div></div>
    <div class="info-card glass"><div class="label">Mobile Banking</div><div class="value" style="font-size:15px;color:${u.bank
ing.mobileBanking?'#5fe0a0':'#ff8a8a'}">${u.banking.mobileBanking?'Active':'Inactive'}</div></div>
    <div class="info-card glass"><div class="label">Debit Card</div><div class="value" style="font-size:15px;color:${u.banking.
debitCard?'#5fe0a0':'#ff8a8a'}">${u.banking.debitCard?'Issued':'Not Issued'}</div></div>
    <div class="info-card glass"><div class="label">Credit Card</div><div class="value" style="font-size:15px;color:${u.bankin
g.creditCard?'#5fe0a0':'#ff8a8a'}">${u.banking.creditCard?'Issued':'Not Issued'}</div></div>
    <div class="info-card glass"><div class="label">KYC Status</div><div class="value" style="font-size:15px;color:#5fe0a0;">
${u.account.kycStatus}</div></div>
    <div class="info-card glass"><div class="label">Nominee Status</div><div class="value" style="font-size:15px;color:${u.nomi
nee.nomineeName?'#5fe0a0':'#ff8a8a'}">${u.nominee.nomineeName?'Added: '+u.nominee.nomineeName:'Not Added'}</div></div>
    <div class="info-card glass"><div class="label">Account Status</div><div class="value" style="font-size:15px;color:#5fe0a
0;">${u.account.accountStatus}</div></div>
    <div class="info-card glass"><div class="label">Opening Date</div><div class="value" style="font-size:15px;">${fmtDate(u.ac
count.openingDate)}</div></div>
  `;

  document.getElementById('actionGrid').innerHTML = ACTIONS.map(a=>`
    <div class="action-card glass" onclick="handleAction('${a.id}')">
      <div class="action-icon">${a.icon}</div><span>${a.label}</span>
    </div>`).join('');

  renderRecentTx();
}

function renderRecentTx(){
  const txns = [...currentUser.transactions].sort((a,b)=>new Date(b.date)-new Date(a.date)).slice(0,5);
  document.getElementById('recentTxBody').innerHTML = txns.map(t=>`
    <tr>
      <td>${fmtDate(t.date)}</td>
      <td>${t.description}</td>
      <td class="credit">${t.credit? fmtMoney(t.credit):'—'}</td>
      <td class="debit">${t.debit? fmtMoney(t.debit):'—'}</td>
      <td>${fmtMoney(t.balance)}</td>
    </tr>`).join('');
}

function handleAction(id){
  const map = {
    'view-profile': showViewProfile,
    'edit-profile': showEditProfile,
    'deposit': ()=>showMoneyModal('deposit'),
    'withdraw': ()=>showMoneyModal('withdraw'),
    'transfer': showTransferModal,
    'transactions': showTransactionsModal,
    'passbook': downloadPassbook,
    'card': showCardModal,
    'loan': showLoanModal,
    'fd': showFDModal,
    'settings': showSettingsModal,
    'logout': showLogoutModal
  };
  if(map[id]) map[id]();
}

/* ====================================================
   MODALS — profile
==================================================== */
function showViewProfile(){
  const u = currentUser;
  const rows = (obj, keys)=>keys.map(k=>`<div class="detail-row"><span>${k.label}</span><span>${obj[k.id] || '—'}</span></div>
`).join('');
  const html = modalShell('Your Profile', `
    <h3 style="color:var(--gold-light);font-size:14px;margin:14px 0 6px;">Personal</h3>
    ${rows(u.personal, FIELDS.personal.map(f=>({id:f.id,label:f.label})))}
    <h3 style="color:var(--gold-light);font-size:14px;margin:18px 0 6px;">Contact</h3>
    ${rows(u.contact, FIELDS.contact.map(f=>({id:f.id,label:f.label})))}
    <h3 style="color:var(--gold-light);font-size:14px;margin:18px 0 6px;">Employment</h3>
    ${rows(u.employment, FIELDS.employment.map(f=>({id:f.id,label:f.label})))}
    <h3 style="color:var(--gold-light);font-size:14px;margin:18px 0 6px;">Nominee</h3>
    ${rows(u.nominee, FIELDS.nominee.map(f=>({id:f.id,label:f.label})))}
    <button class="btn btn-gold btn-block" style="margin-top:20px;" onclick="showEditProfile()">Edit Profile</button>
  `);
  openModal(html);
}

function showEditProfile(){
  const u = currentUser;
  const editable = [...FIELDS.personal, ...FIELDS.contact, ...FIELDS.employment].filter(f=>!['aadhaar','pan'].includes(f.id));
  const html = modalShell('Edit Profile', `
    <div style="max-height:56vh;overflow-y:auto;padding-right:6px;">
    ${editable.map(f=>{
      const src = FIELDS.personal.includes(f)?u.personal:(FIELDS.contact.includes(f)?u.contact:u.employment);
      if(f.type === 'select'){
        return `<div class="field"><label>${f.label}</label><select id="ep_${f.id}">${f.opts.map(o=>`<option value="${o}" ${src
[f.id]===o?'selected':''}>${o}</option>`).join('')}</select></div>`;
      }
      return `<div class="field"><label>${f.label}</label><input type="${f.type}" id="ep_${f.id}" value="${src[f.id]||''}"></di
v>`;
```

```
    }).join('')}
    </div>
    <button class="btn btn-gold btn-block" style="margin-top:14px;" onclick="saveProfile()">Save Changes</button>
  `);
  openModal(html);
}

function saveProfile(){
  const u = currentUser;
  const editable = [...FIELDS.personal, ...FIELDS.contact, ...FIELDS.employment].filter(f=>!['aadhaar','pan'].includes(f.id));
  editable.forEach(f=>{
    const el = document.getElementById('ep_'+f.id);
    if(!el) return;
    const val = el.value.trim();
    if(FIELDS.personal.includes(f)) u.personal[f.id]=val;
    else if(FIELDS.contact.includes(f)) u.contact[f.id]=val;
    else u.employment[f.id]=val;
  });
  DB.saveUser(u);
  renderDashboard();
  showToast('Profile updated successfully');
  closeModal();
}

/* ================================================
   MODALS — money movement
================================================ */
function addTransaction(desc, credit, debit){
  const u = currentUser;
  u.account.balance += (credit - debit);
  u.account.ledgerBalance = u.account.balance;
  u.transactions.push({
    id:'TXN'+randDigits(10), date:new Date().toISOString(), description:desc,
    credit, debit, balance:u.account.balance, status:'Success'
  });
  DB.saveUser(u);
}

function showMoneyModal(kind){
  const isDeposit = kind==='deposit';
  const html = modalShell(isDeposit?'Deposit Money':'Withdraw Money', `
    <p style="color:var(--text-muted);font-size:13.5px;margin-bottom:16px;">Available Balance: <strong style="color:var(--gold-
light)">${fmtMoney(currentUser.account.balance)}</strong></p>
    <div class="field"><label>Amount (₹)</label><input type="number" id="moneyAmt" placeholder="Enter amount" min="1"></div>
    <div class="field"><label>Remarks</label><input type="text" id="moneyRemark" placeholder="Optional note" value="${isDeposi
t?'Cash Deposit':'Cash Withdrawal'}"></div>
    <button class="btn btn-gold btn-block" onclick="submitMoney('${kind}')">${isDeposit?'Deposit':'Withdraw'}</button>
  `);
  openModal(html);
}
function submitMoney(kind){
  const amt = Number(document.getElementById('moneyAmt').value);
  const remark = document.getElementById('moneyRemark').value.trim() || (kind==='deposit'?'Cash Deposit':'Cash Withdrawal');
  if(!amt || amt <= 0){ showToast('Enter a valid amount'); return; }
  if(kind === 'withdraw' && amt > currentUser.account.balance){ showToast('Insufficient balance'); return; }
  addTransaction(remark, kind==='deposit'?amt:0, kind==='withdraw'?amt:0);
  renderDashboard();
  showToast(`${kind==='deposit'?'Deposited':'Withdrew'} ${fmtMoney(amt)} successfully`);
  closeModal();
}

function showTransferModal(){
  const html = modalShell('Transfer Money', `
    <p style="color:var(--text-muted);font-size:13.5px;margin-bottom:16px;">Available Balance: <strong style="color:var(--gold-
light)">${fmtMoney(currentUser.account.balance)}</strong></p>
    <div class="field"><label>Recipient Account Number</label><input type="text" id="transferAcc" placeholder="Enter 16-digit a
ccount number" maxlength="16"></div>
    <div class="field"><label>Recipient Name</label><input type="text" id="transferName" placeholder="Enter recipient name"></d
iv>
    <div class="field"><label>Amount (₹)</label><input type="number" id="transferAmt" placeholder="Enter amount"></div>
    <div class="field"><label>Transaction PIN</label><input type="password" id="transferPin" placeholder="Enter your 4-digit PI
N" maxlength="6"></div>
    <button class="btn btn-gold btn-block" onclick="submitTransfer()">Transfer</button>
  `);
  openModal(html);
}
function submitTransfer(){
  const acc = document.getElementById('transferAcc').value.trim();
  const name = document.getElementById('transferName').value.trim();
  const amt = Number(document.getElementById('transferAmt').value);
  const pin = document.getElementById('transferPin').value.trim();
  if(!acc || !/^\d{6,20}$/.test(acc)){ showToast('Enter a valid numeric account number'); return; }
  if(!name || !PATTERNS.name.test(name)){ showToast('Enter a valid recipient name (letters only)'); return; }
  if(!amt || amt <= 0){ showToast('Enter a valid amount'); return; }
  if(amt > currentUser.account.balance){ showToast('Insufficient balance'); return; }
  if(pin !== currentUser.security.transactionPin){ showToast('Incorrect transaction PIN'); return; }
  addTransaction(`UPI Transfer to ${name} (${acc.slice(-4)})`, 0, amt);
  renderDashboard();
  showToast(`Transferred ${fmtMoney(amt)} to ${name}`);
  closeModal();
```

```
}

/* ====================================================
   MODALS — transaction history (search + filter)
==================================================== */
function showTransactionsModal(){
  const html = modalShell('Transaction History', `
    <div class="filter-bar">
      <input type="text" id="txSearch" placeholder="Search description or ID" oninput="renderTxTable()">
      <select id="txFilter" onchange="renderTxTable()">
        <option value="all">All</option>
        <option value="credit">Credits Only</option>
        <option value="debit">Debits Only</option>
      </select>
    </div>
    <div class="table-wrap" style="max-height:50vh;overflow-y:auto;">
      <table>
        <thead><tr><th>Date</th><th>Time</th><th>Description</th><th>Credit</th><th>Debit</th><th>Balance</th><th>Txn ID</th><t
h>Status</th></tr></thead>
        <tbody id="txTableBody"></tbody>
      </table>
    </div>
  `, 'modal-wide');
  openModal(html);
  renderTxTable();
}
function renderTxTable(){
  const search = (document.getElementById('txSearch')?.value || '').toLowerCase();
  const filter = document.getElementById('txFilter')?.value || 'all';
  let txns = [...currentUser.transactions].sort((a,b)=>new Date(b.date)-new Date(a.date));
  if(filter==='credit') txns = txns.filter(t=>t.credit>0);
  if(filter==='debit') txns = txns.filter(t=>t.debit>0);
  if(search) txns = txns.filter(t=> t.description.toLowerCase().includes(search) || t.id.toLowerCase().includes(search));
  document.getElementById('txTableBody').innerHTML = txns.map(t=>{
    const dt = new Date(t.date);
    return `<tr>
      <td>${fmtDate(t.date)}</td>
      <td>${dt.toLocaleTimeString('en-IN',{hour:'2-digit',minute:'2-digit'})}</td>
      <td>${t.description}</td>
      <td class="credit">${t.credit? fmtMoney(t.credit):'—'}</td>
      <td class="debit">${t.debit? fmtMoney(t.debit):'—'}</td>
      <td>${fmtMoney(t.balance)}</td>
      <td style="font-size:11px;color:var(--text-muted);">${t.id}</td>
      <td><span class="badge success">${t.status}</span></td>
    </tr>`;
  }).join('') || `<tr><td colspan="8" style="text-align:center;color:var(--text-muted);padding:24px;">No transactions found</td
></tr>`;
}

function downloadPassbook(){
  const u = currentUser;
  let content = `VAULTRA BANK — ACCOUNT PASSBOOK\n`;
  content += `================================\n`;
  content += `Customer: ${u.personal.fullName}\nAccount No: ${u.account.accountNumber}\nIFSC: ${u.account.ifsc}\nBranch: ${u.ac
count.branch}\n\n`;
  content += `Date        Description                              Credit        Debit         Balance\n`;
  content += `-------------------------------------------------------------------------------------\n`;
  [...u.transactions].sort((a,b)=>new Date(a.date)-new Date(b.date)).forEach(t=>{
    content += `${fmtDate(t.date).padEnd(12)}${t.description.slice(0,38).padEnd(41)}${(t.credit?fmtMoney(t.credit):'-').padEnd
(14)}${(t.debit?fmtMoney(t.debit):'-').padEnd(14)}${fmtMoney(t.balance)}\n`;
  });
  const blob = new Blob([content], {type:'text/plain'});
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href = url; a.download = `Vaultra_Passbook_${u.account.accountNumber}.txt`;
  document.body.appendChild(a); a.click(); a.remove();
  URL.revokeObjectURL(url);
  showToast('Passbook downloaded');
}

/* ====================================================
   MODALS — card, loan, FD, settings, logout
==================================================== */
function showCardModal(){
  const u = currentUser;
  const html = modalShell('Card Details', `
    <div class="vault-card" style="width:100%;height:190px;margin-bottom:22px;">
      <div class="vault-chip"></div>
      <div class="vault-num">${u.banking.debitCard ? u.account.cardNumber : '•••• •••• •••• ••••'}</div>
      <div class="vault-foot"><span>${(u.personal.fullName||'CARD HOLDER').toUpperCase()}</span><span>VALID THRU ${u.account.ca
rdExpiry}</span></div>
    </div>
    <div class="detail-row"><span>Debit Card</span><span>${u.banking.debitCard?'Issued & Active':'Not Requested'}</span></div>
    <div class="detail-row"><span>Credit Card</span><span>${u.banking.creditCard?'Issued & Active':'Not Requested'}</span></div
>
    <div class="detail-row"><span>ATM Facility</span><span>${u.banking.atmFacility?'Enabled':'Disabled'}</span></div>
    <div class="detail-row"><span>Daily ATM Limit</span><span>${fmtMoney(50000)}</span></div>
    <div class="detail-row"><span>Daily POS Limit</span><span>${fmtMoney(100000)}</span></div>
  `);
  openModal(html);
```

```
}

function showLoanModal(){
  const income = Number(currentUser.employment.monthlyIncome || 0);
  const html = modalShell('Loan Eligibility', `
    <p style="color:var(--text-muted);font-size:13.5px;margin-bottom:16px;">Based on your monthly income of <strong style="colo
r:var(--gold-light)">${fmtMoney(income)}</strong>, here's your estimated eligibility:</p>
    <div class="field"><label>Loan Type</label>
      <select id="loanType" onchange="calcLoan()">
        <option value="0.5">Personal Loan (up to 50× monthly income)</option>
        <option value="4">Home Loan (up to 400× monthly income)</option>
        <option value="2">Car Loan (up to 200× monthly income)</option>
      </select>
    </div>
    <div class="detail-row"><span>Estimated Eligible Amount</span><span id="loanEligible" style="color:var(--gold-light);font-s
ize:16px;">—</span></div>
    <div class="detail-row"><span>Est. Interest Rate</span><span id="loanRate">—</span></div>
    <div class="detail-row"><span>Max Tenure</span><span id="loanTenure">—</span></div>
    <p style="font-size:12px;color:var(--text-muted);margin-top:14px;">Indicative estimate only — final approval subject to cre
dit assessment.</p>
  `);
  openModal(html);
  calcLoan();
}
function calcLoan(){
  const income = Number(currentUser.employment.monthlyIncome || 0);
  const mult = Number(document.getElementById('loanType').value);
  const rates = {0.5:'11.5% p.a.', 4:'8.4% p.a.', 2:'9.2% p.a.'};
  const tenures = {0.5:'5 years', 4:'20 years', 2:'7 years'};
  document.getElementById('loanEligible').textContent = fmtMoney(income * mult * 50);
  document.getElementById('loanRate').textContent = rates[mult];
  document.getElementById('loanTenure').textContent = tenures[mult];
}

function showFDModal(){
  const html = modalShell('Fixed Deposit', `
    <p style="color:var(--text-muted);font-size:13.5px;margin-bottom:16px;">Available Balance: <strong style="color:var(--gold-
light)">${fmtMoney(currentUser.account.balance)}</strong></p>
    <div class="field"><label>Deposit Amount (₹)</label><input type="number" id="fdAmt" placeholder="Minimum ₹1,000" oninput="c
alcFD()"></div>
    <div class="field"><label>Tenure</label>
      <select id="fdTenure" onchange="calcFD()">
        <option value="0.5">6 Months (6.0% p.a.)</option>
        <option value="1">1 Year (6.8% p.a.)</option>
        <option value="3">3 Years (7.2% p.a.)</option>
        <option value="5">5 Years (7.5% p.a.)</option>
      </select>
    </div>
    <div class="detail-row"><span>Maturity Amount</span><span id="fdMaturity" style="color:var(--gold-light);">—</span></div>
    <div class="detail-row"><span>Interest Earned</span><span id="fdInterest">—</span></div>
    <button class="btn btn-gold btn-block" style="margin-top:16px;" onclick="createFD()">Book Fixed Deposit</button>
  `);
  openModal(html);
}
function calcFD(){
  const amt = Number(document.getElementById('fdAmt').value || 0);
  const yrs = Number(document.getElementById('fdTenure').value);
  const rates = {0.5:0.06, 1:0.068, 3:0.072, 5:0.075};
  const rate = rates[yrs];
  const maturity = amt * Math.pow(1+rate, yrs);
  document.getElementById('fdMaturity').textContent = fmtMoney(maturity||0);
  document.getElementById('fdInterest').textContent = fmtMoney((maturity-amt)||0);
}
function createFD(){
  const amt = Number(document.getElementById('fdAmt').value || 0);
  if(!amt || amt < 1000){ showToast('Minimum FD amount is ₹1,000'); return; }
  if(amt > currentUser.account.balance){ showToast('Insufficient balance'); return; }
  const yrs = Number(document.getElementById('fdTenure').value);
  addTransaction(`Fixed Deposit Booked (${yrs} yr)`, 0, amt);
  renderDashboard();
  showToast(`Fixed Deposit of ${fmtMoney(amt)} booked successfully`);
  closeModal();
}

function showSettingsModal(){
  const html = modalShell('Settings', `
    <h3 style="color:var(--gold-light);font-size:14px;margin-bottom:10px;">Change Password</h3>
    <div class="field pw-wrap"><label>Current Password</label><input type="password" id="setCurPw"></div>
    <div class="field pw-wrap"><label>New Password</label><input type="password" id="setNewPw" placeholder="Min 8 chars, 1 uppe
r, 1 number, 1 symbol"></div>
    <button class="btn btn-royal btn-block" onclick="changePassword()">Update Password</button>
    <h3 style="color:var(--gold-light);font-size:14px;margin:22px 0 10px;">Notification Preferences</h3>
    <div class="toggle-item" style="margin-bottom:10px;"><span>SMS Alerts</span><label class="switch"><input type="checkbox" id
="setSms" ${currentUser.banking.smsAlerts?'checked':''} onchange="toggleSetting('smsAlerts')"><span class="slider-toggle"></spa
n></label></div>
    <div class="toggle-item"><span>Internet Banking</span><label class="switch"><input type="checkbox" id="setIb" ${currentUse
r.banking.internetBanking?'checked':''} onchange="toggleSetting('internetBanking')"><span class="slider-toggle"></span></label>
</div>
  `);
  openModal(html);
```

```
}
function changePassword(){
  const cur = document.getElementById('setCurPw').value;
  const next = document.getElementById('setNewPw').value;
  if(cur !== currentUser.security.password){ showToast('Current password is incorrect'); return; }
  if(!PATTERNS.password.test(next)){ showToast('New password does not meet strength requirements'); return; }
  currentUser.security.password = next;
  DB.saveUser(currentUser);
  showToast('Password updated successfully');
  closeModal();
}
function toggleSetting(key){
  currentUser.banking[key] = !currentUser.banking[key];
  DB.saveUser(currentUser);
  renderDashboard();
}

function showLogoutModal(){
  const html = modalShell('Confirm Logout', `
    <p style="color:var(--text-muted);font-size:14px;margin-bottom:22px;">Are you sure you want to log out of your Vaultra Bank
account?</p>
    <div style="display:flex;gap:12px;">
      <button class="btn btn-outline" style="flex:1;" onclick="closeModal()">Cancel</button>
      <button class="btn btn-danger" style="flex:1;" onclick="doLogout()">Logout</button>
    </div>
  `);
  openModal(html);
}
function doLogout(){
  DB.clearCurrent();
  currentUser = null;
  closeModal();
  goTo('page-login');
  document.getElementById('loginForm').reset();
  showToast('You have been logged out');
}

/* ======================================================
   INIT
====================================================== */
function initApp(){
  renderStepLabels();
  renderSteps();
  updateProgress();

  const existing = DB.getCurrent();
  if(existing && DB.getUser(existing)){
    loadDashboard(existing);
    document.querySelectorAll('.page').forEach(p=>p.classList.remove('active'));
    document.getElementById('page-dashboard').classList.add('active');
  }

  document.getElementById('regForm').addEventListener('submit', e=>e.preventDefault());
}
initApp();
</script>
</body>
</html>
```

## 10. Conclusion

All modules were joined into one working project. The portal lets a customer register with validated details, receive account numbers and a virtual card, log in securely, and perform banking operations with every action recorded in the transaction history.